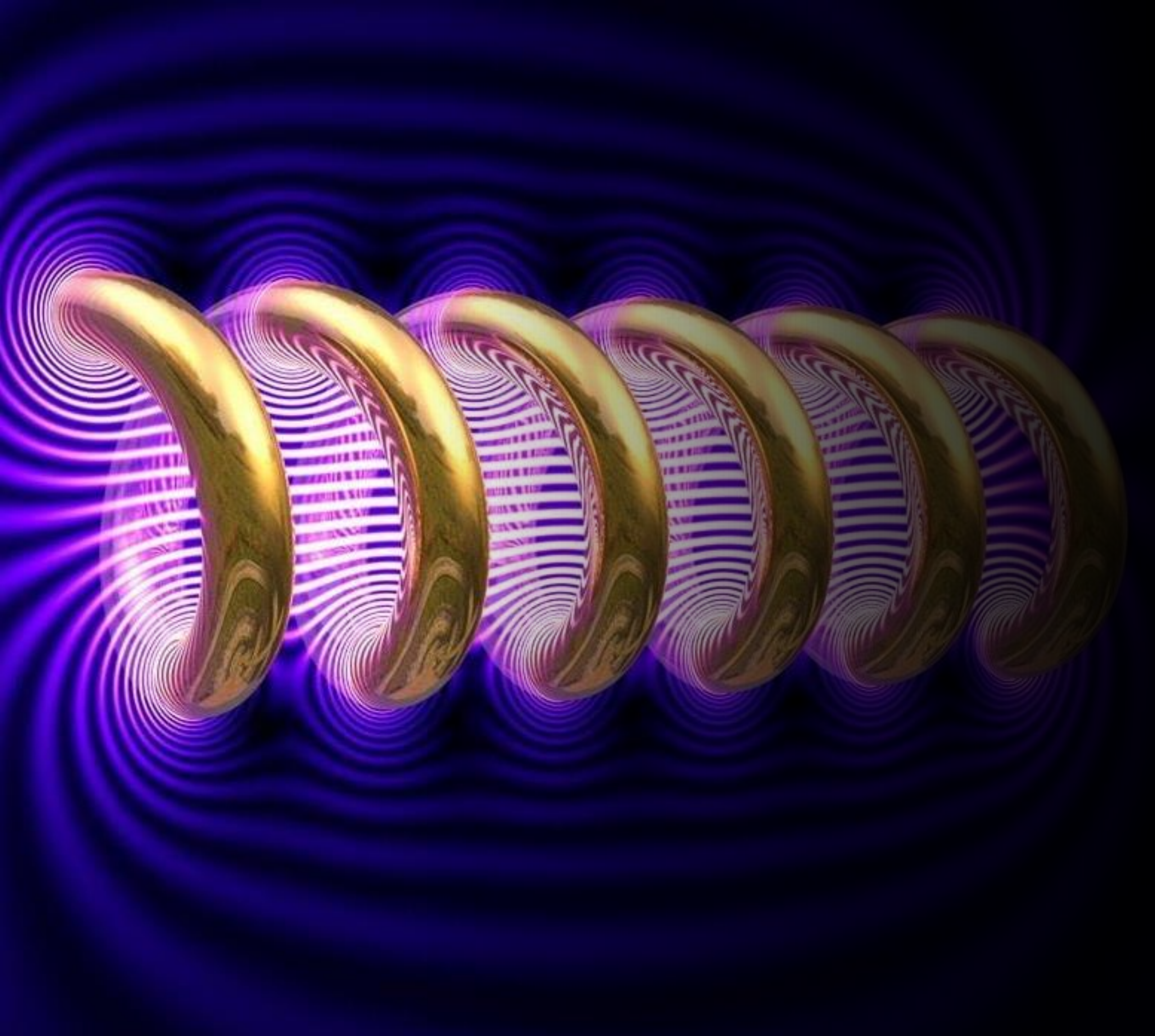
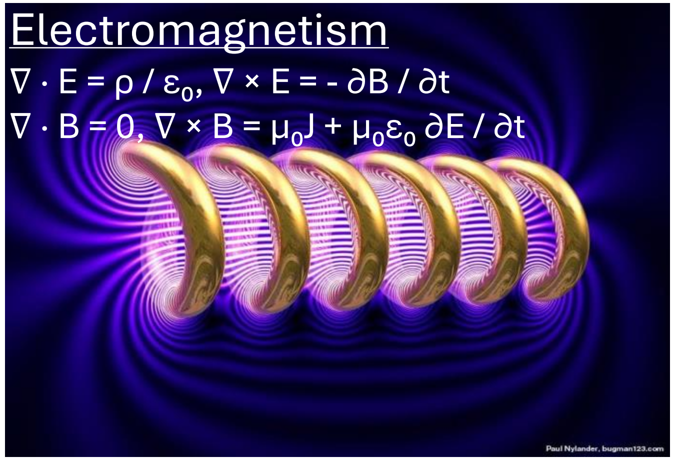


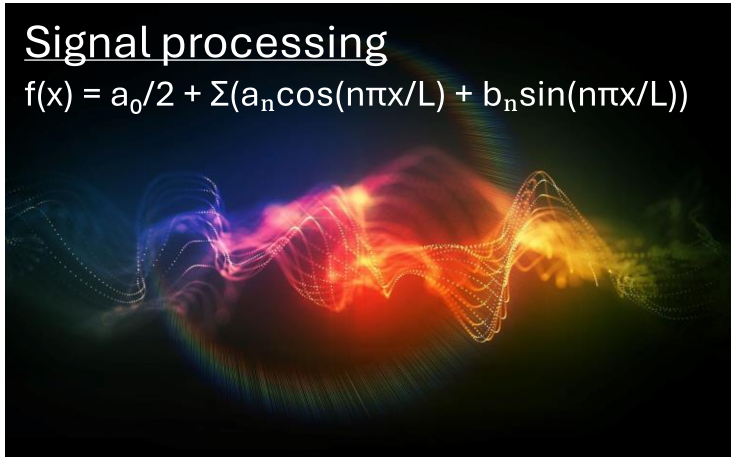
Mathematics 2 (5EZB0)
Lecture 1

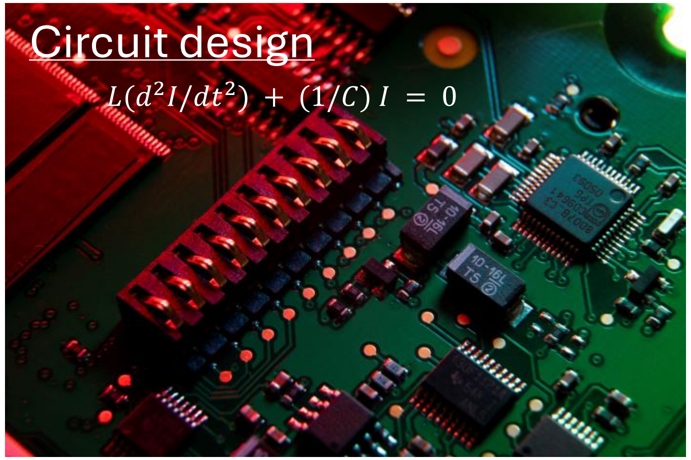
Koondi Mitra

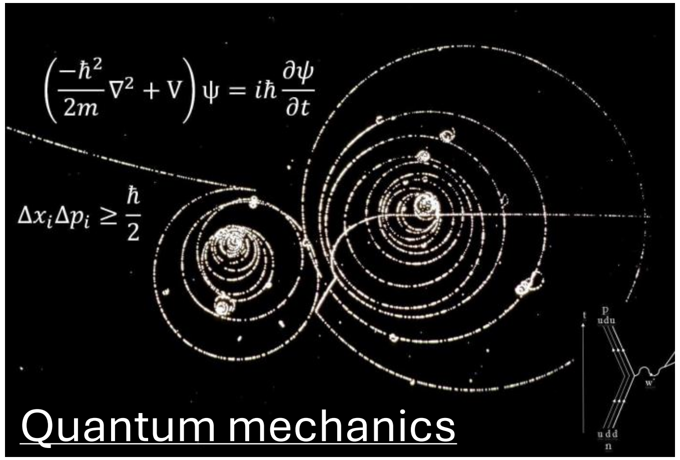
Mathematics & Computer Science Department

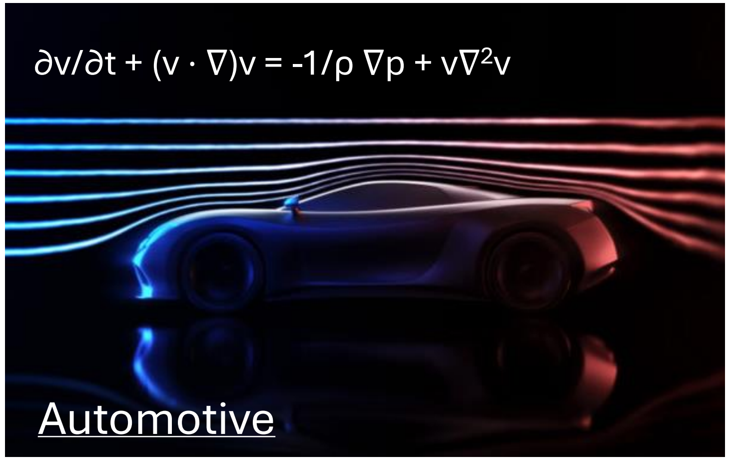
Multivariable calculus

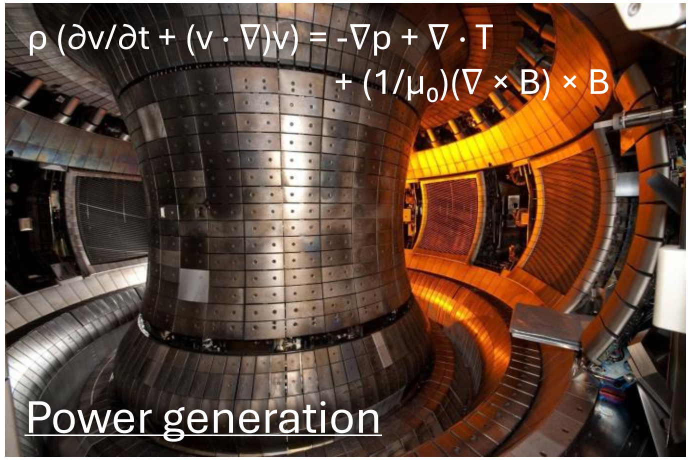












Moving to 3D

Let us first consider Cartesian coordinate system with x, y, z axes

- They have to be oriented by the right-handed rule

- They need to satisfy

  $\boldsymbol{i} \times \boldsymbol{j} = \boldsymbol{k}$

  $\boldsymbol{k} \times \boldsymbol{i} = \boldsymbol{j}$

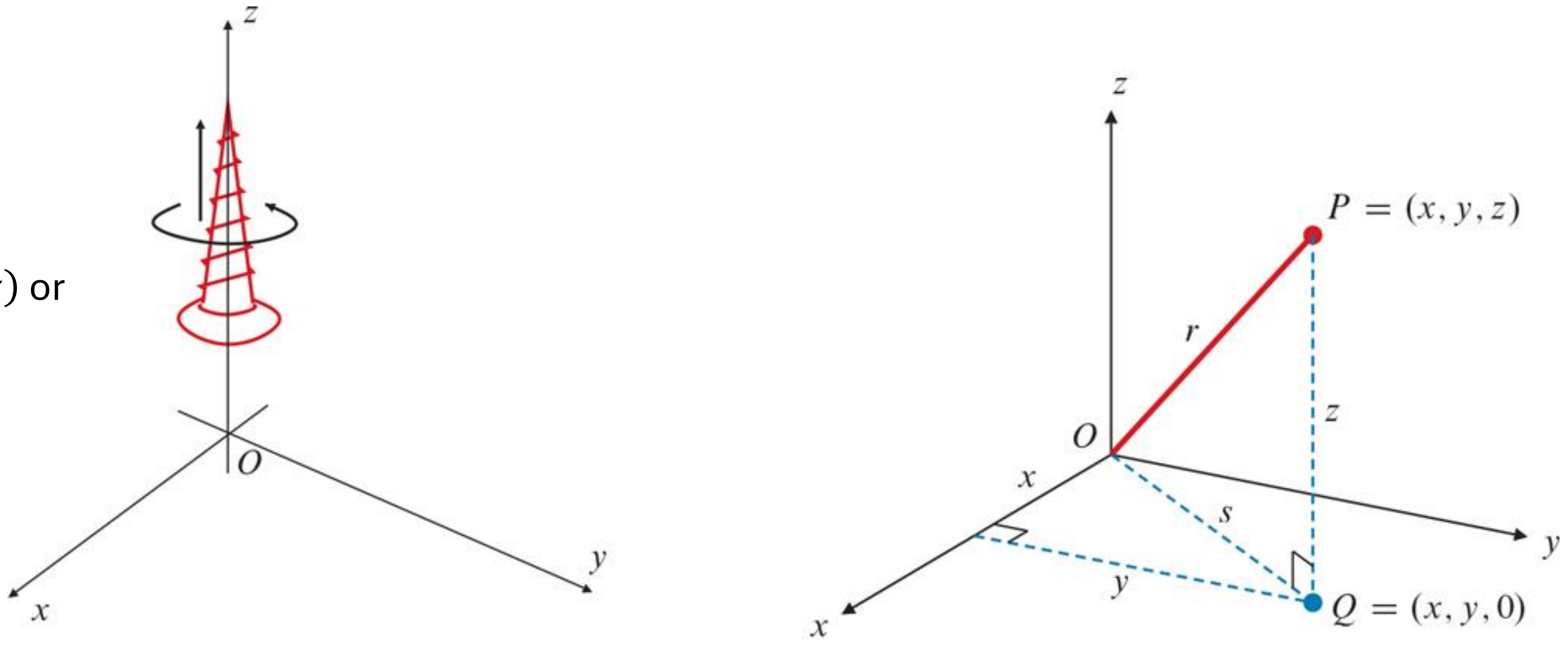
  $\boldsymbol{j} \times \boldsymbol{k} = \boldsymbol{i}$

- The coordinate is $\mathrm{P} = (x, y, z)$ or

  $\boldsymbol{x} = x\,\boldsymbol{i} + y\,\boldsymbol{j} + z\,\boldsymbol{k}$

- Distance from the origin

  $|\,\boldsymbol{x}\,| = \sqrt{x^2 + y^2 + z^2}$



(a) (b)

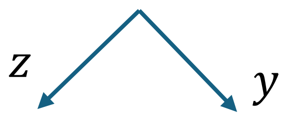
Euclidean distance

- Distance between 2 points $\mathrm{P}_1 = (x_1, y_1, z_1)$ and $\mathrm{P}_2 = (x_2, y_2, z_2)$:

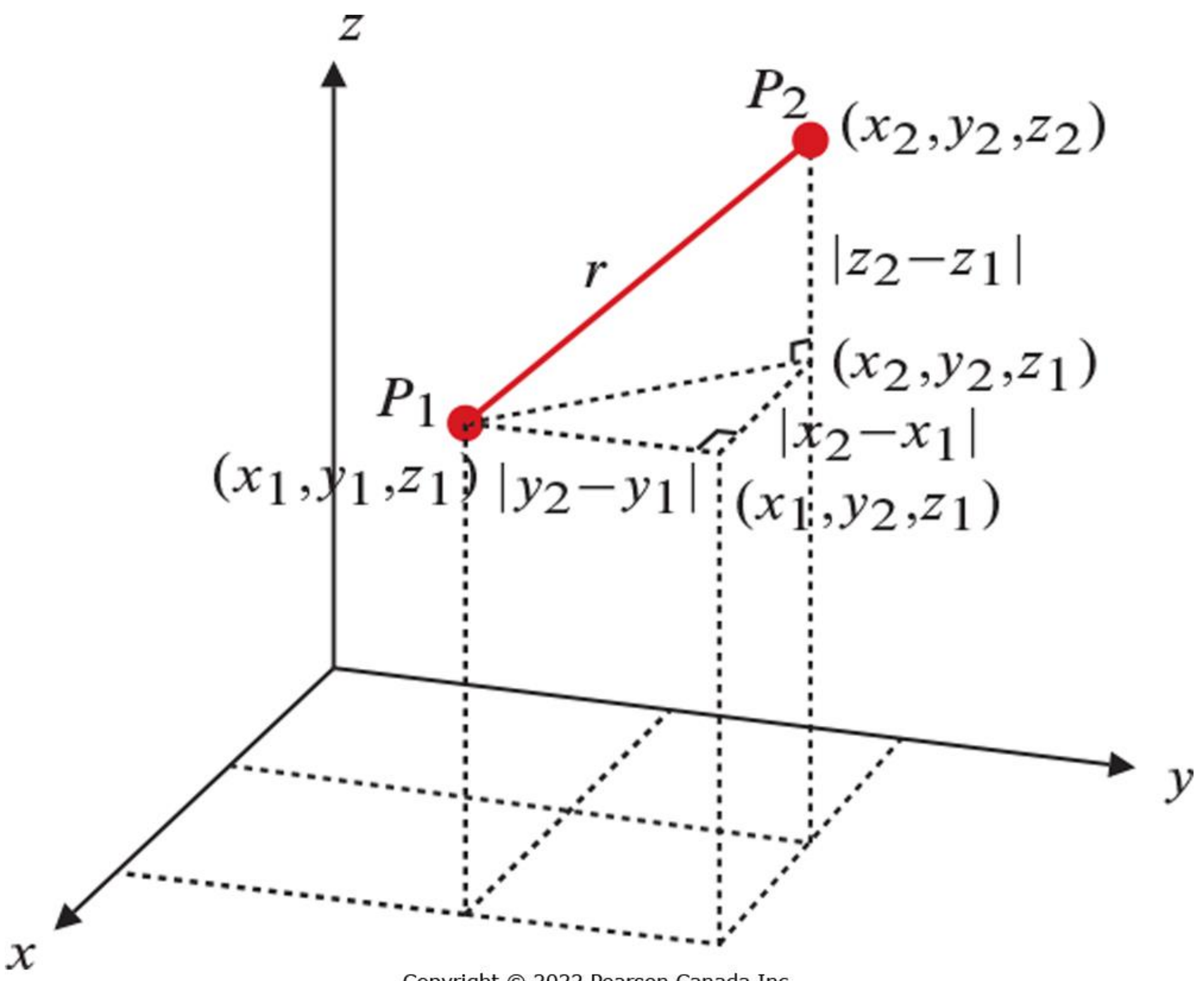
$$|\boldsymbol{x_2} - \boldsymbol{x_1}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Questions

1. What is the orientation of x-axis in the picture below?



2. What is the length of $(3,4,12)$ and what is the angle it makes with the x-axis?

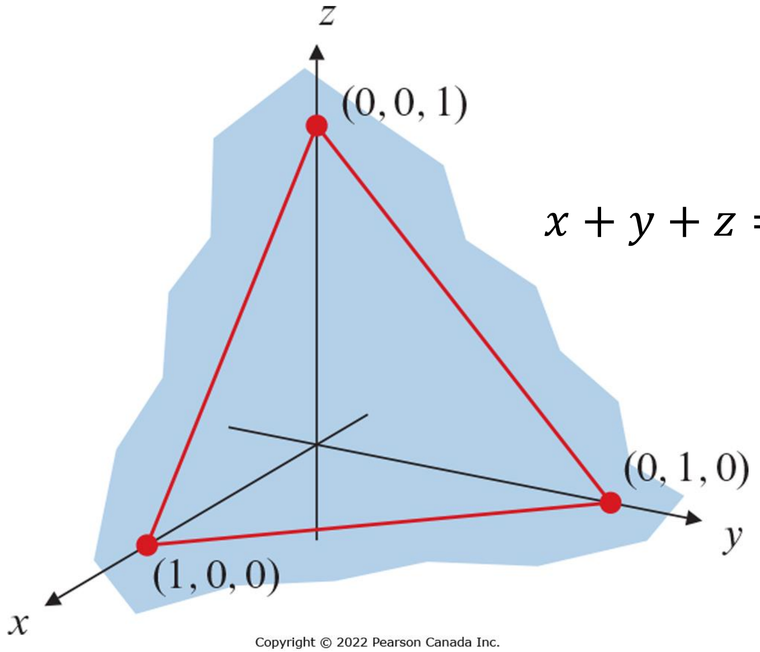


Equation of planes

- The equation of a plane is for constants $a, b, c, d \in \mathbb{R}$,
$$ax + by + cz = d$$

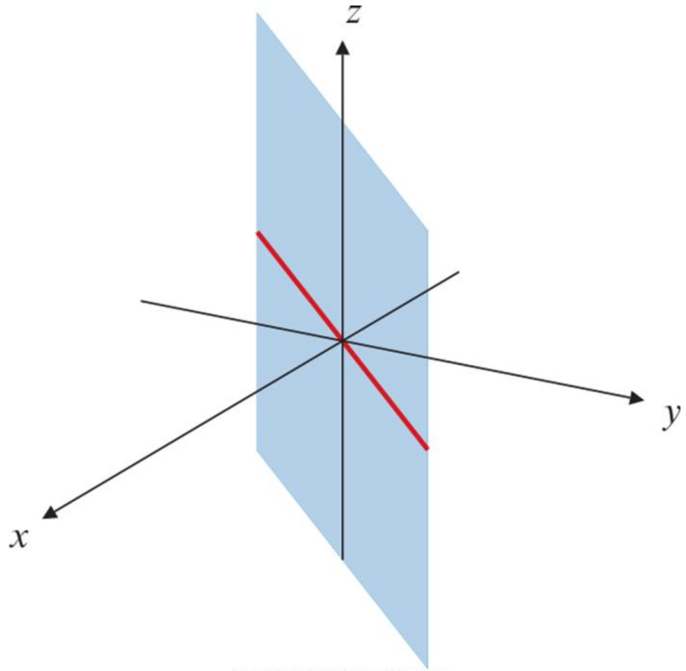
To see how the plane changes for $c > 0$ play with the python code **plane.py**

or if $\boldsymbol{n} = a\boldsymbol{i} + b\boldsymbol{j} + c\boldsymbol{k}$, then $\boldsymbol{n} \cdot \boldsymbol{x} = d.$



$x + y + z = 1$

$x = y$



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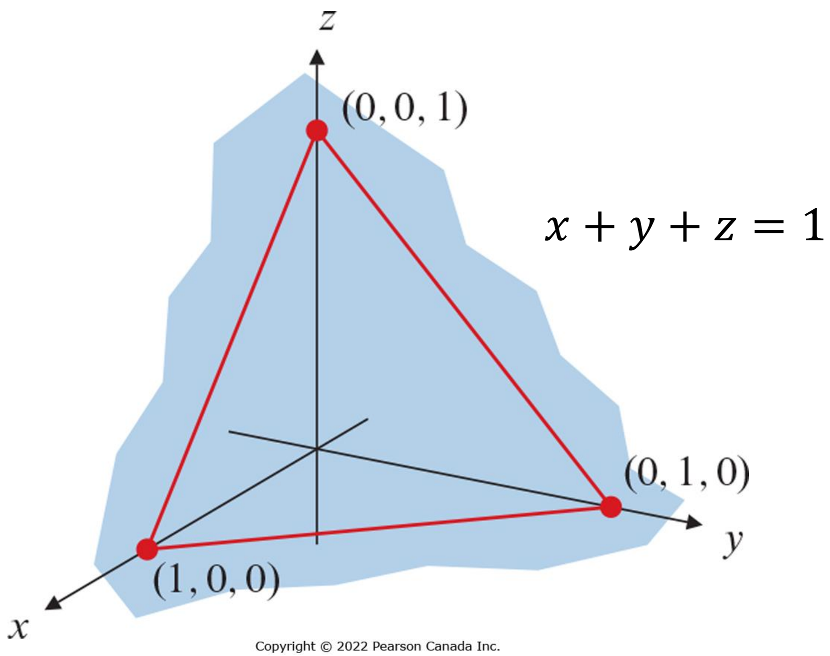


Questions

1. What is the unit normal to the plane $x + 2y + 3z = 4$?

2. What is geometry of the intersection between $x + y + z = 1$ and the plane above?

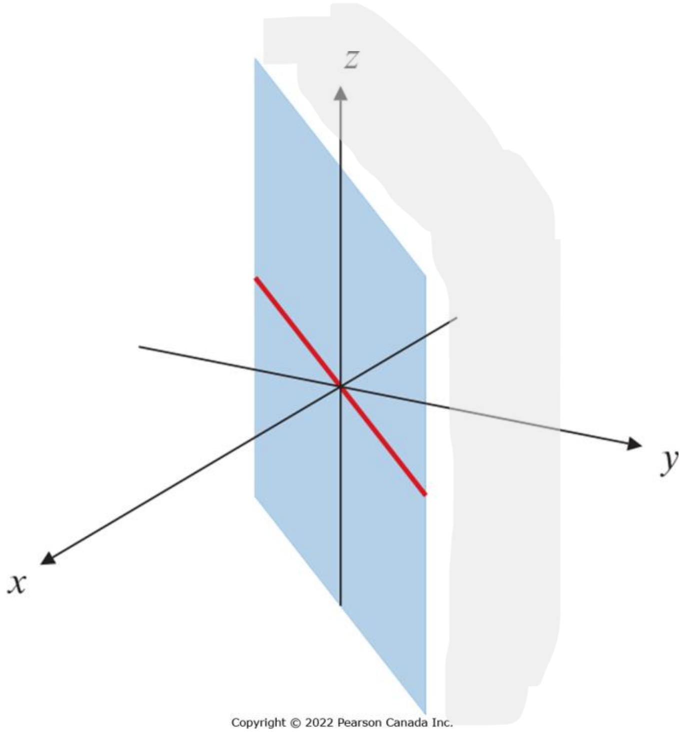
Half spaces

- We get a half space if either

$$ax + by + cz > d$$

or if $$ax + by + cz < d$$

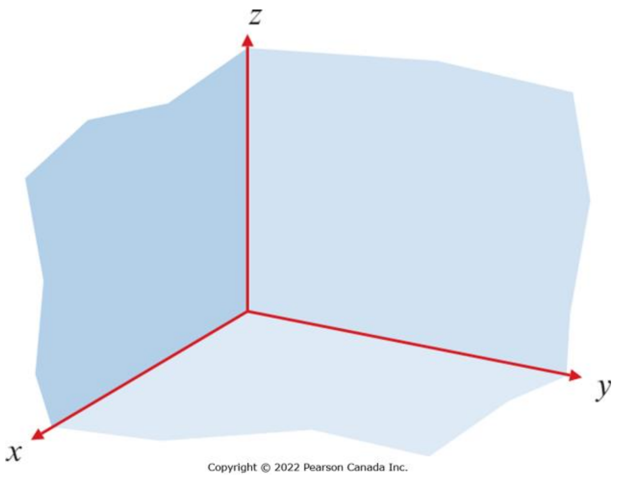
$y > x$



Examples

1. $y > x$ is a half space

2. $\{x > 0, y > 0, z > 0\}$ is an octant which is the intersection of the three half-spaces $x > 0,\ y > 0$ and $z > 0$



Open and closed sets

- We define the ball of radius $r > 0$ centered at $\boldsymbol{x} \in \mathbb{R}^n$ as

$$B_r(\boldsymbol{x}) := \{\boldsymbol{y} \in \mathbb{R}^n : |\boldsymbol{y} - \boldsymbol{x}| < r\}$$

- $S \subseteq \mathbb{R}^n$ is **open** if for all $\boldsymbol{x} \in S$, there exists an $r > 0$ such that

$$B_r(\boldsymbol{x}) \subseteq S$$

- A set S is **closed** if its complement S^c is open.

Questions

1. Is $\{x + 2y + 3z \geq 4\}$ an open or a closed set?

2. Is $\{x^2 + y^2 < 1\}$ open?

3. Is $\{x^2 + y^2 < 1 \ \& \ y \geq x\}$ open or closed?

Quadric surfaces

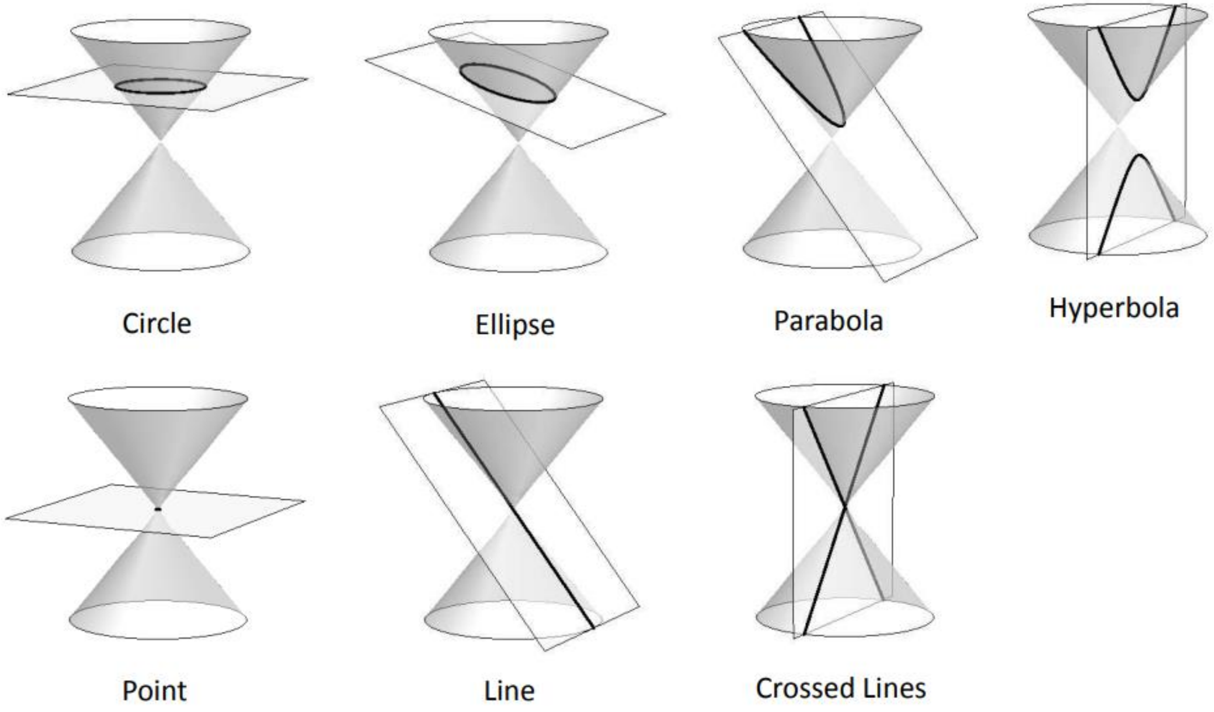
- Next, we discuss quadric surfaces which arise from the equation

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz = J$$

- Observe that the $z = k$ intersection of above equations are conic sections given by

$$Ax^2 + By^2 + Dxy + G'x + H'y = J'$$

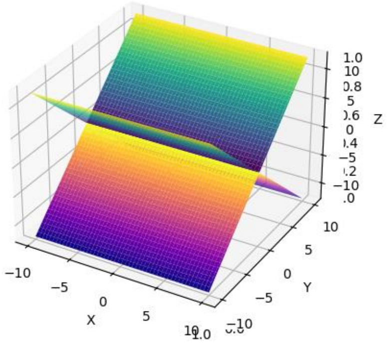
To see how the quadric surfaces change for $c \geq 0$ play with the python code **quadric_surface.py**



Quadric surfaces

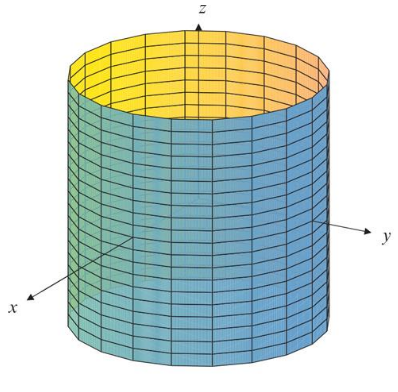
Pair of planes

$$(a_1x + b_1y + c_1z - d_1)(a_1x + b_1y + c_1z - d_2) = 0$$



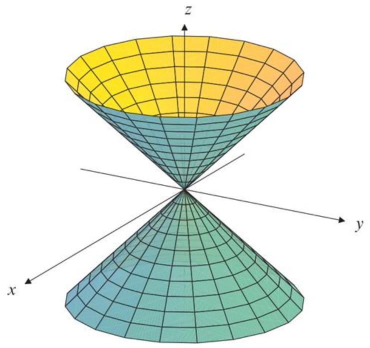
Cylinders along z-axis

$$Ax^2 + By^2 + Dxy + Gx + Hy = J$$



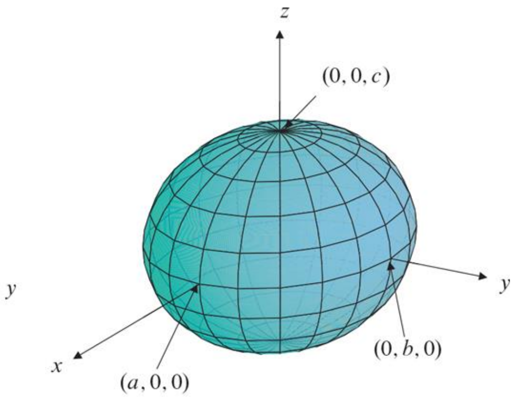
(a)

Cones centered at (x_0, y_0, z_0)



$$\frac{(z - z_0)^2}{c^2} = \frac{(x - x_0)^2}{a^2} + \frac{(y - y_0)^2}{b^2}$$

Ellipsoids centered at (x_0, y_0, z_0)

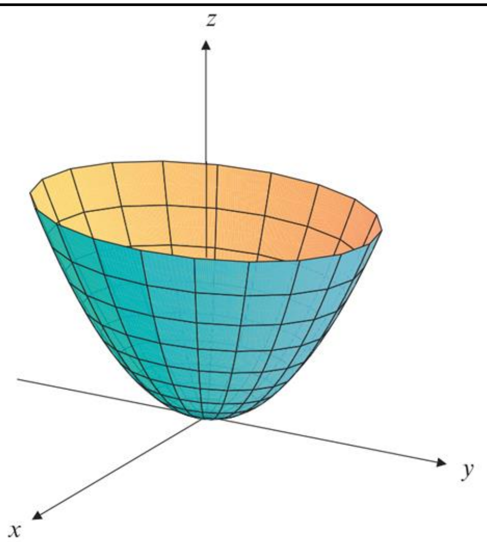


$$\frac{(x - x_0)^2}{a^2} + \frac{(y - y_0)^2}{b^2} + \frac{(z - z_0)^2}{c^2} = 1$$

Quadric surfaces

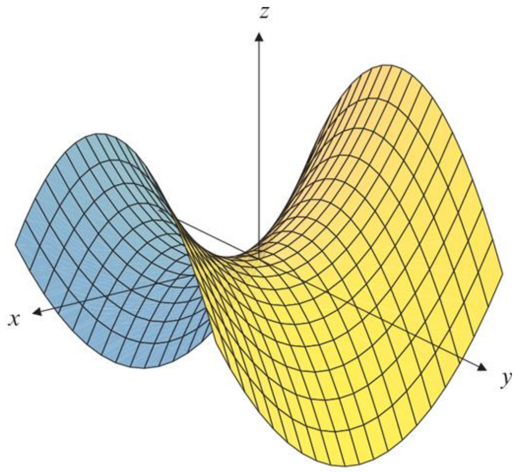
Elliptic paraboloids

$$z - z_0 = \frac{(x - x_0)^2}{a^2} + \frac{(y - y_0)^2}{b^2}$$

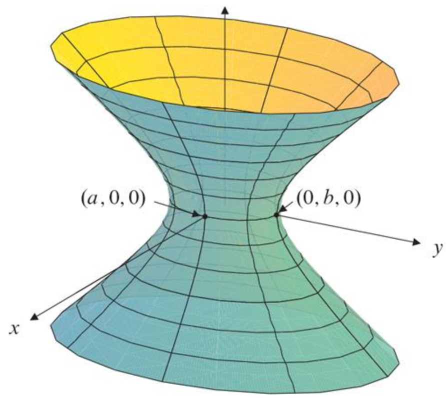


Hyperbolic paraboloids

$$z - z_0 = \frac{(x - x_0)^2}{a^2} - \frac{(y - y_0)^2}{b^2}$$

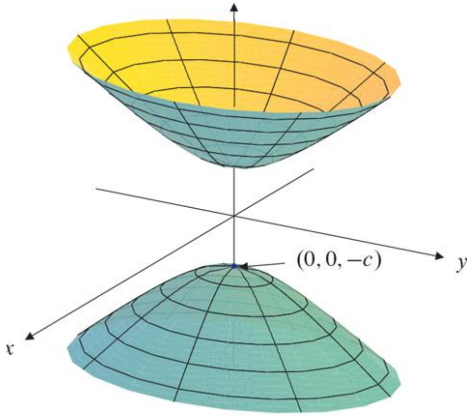


Hyperboloid of one sheet



$$\frac{(z - z_0)^2}{c^2} = \frac{(x - x_0)^2}{a^2} + \frac{(y - y_0)^2}{b^2} - 1$$

Hyperboloid of two sheets



$$\frac{(z - z_0)^2}{c^2} = \frac{(x - x_0)^2}{a^2} + \frac{(y - y_0)^2}{b^2} + 1$$

Quadric surfaces

Questions

1. What kind of surface is $x^2 + y^2 + 2z^2 = 2y - 8z$?

2. What is the equation of a sphere of radius $r > 0$ centered at $(1,2,3)$?

3. What is the relation between conic sections and the quadric surfaces? Analyze the case of hyperboloids.

Questions

4. What is the intersection between $x^2 + 2y^2 + 3z^2 = 6$ and $x = y$?

5. Find a unit vector $\boldsymbol{n}$ perpendicular to which if you intersect the elliptic cone $x^2 + 2y^2 = 1$ then you get a circle.

Cylindrical coordinates

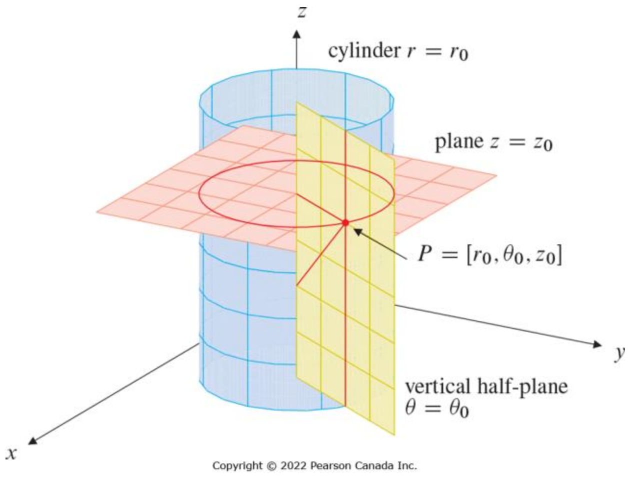
- Useful coordinate system to describe cylindrical objects such as, wire, axle, etc.

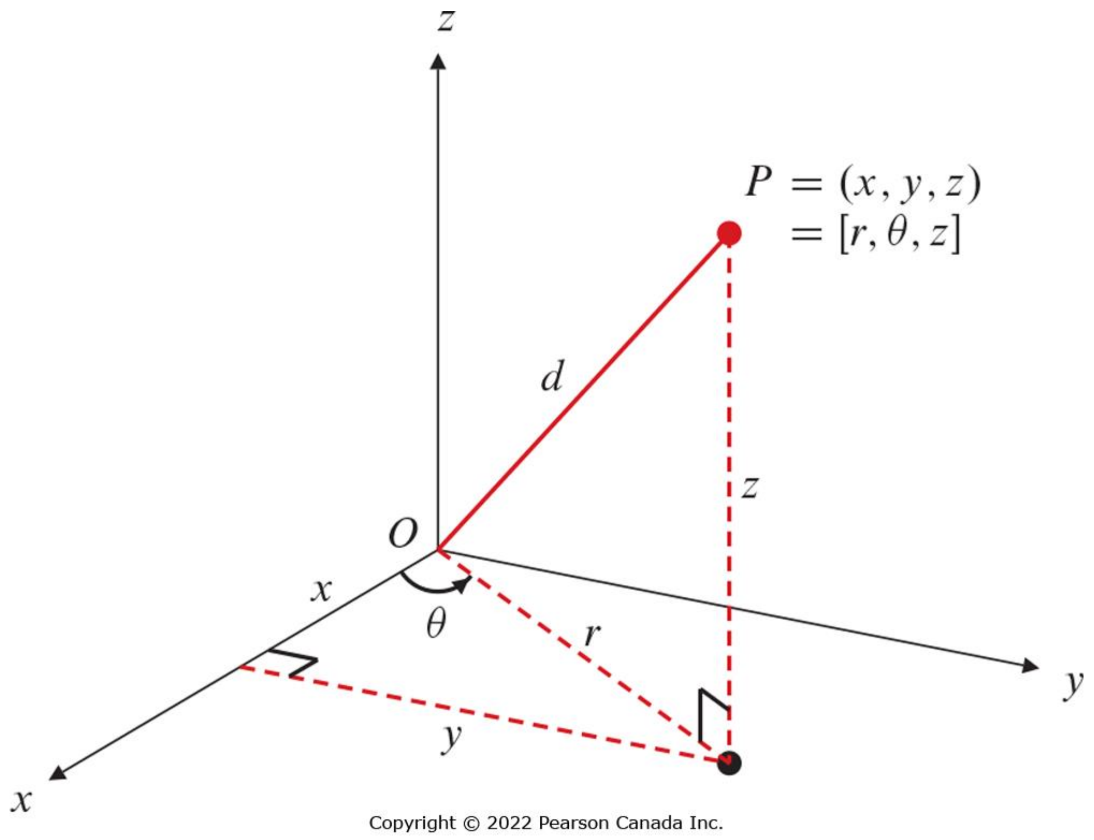
- Represented by $[r, \theta, z]$ where $r \geq 0, z \in \mathbb{R}$ and $\theta \in [-\pi, \pi]$

- The coordinate in in terms of (x, y, z)

$$\boldsymbol{x} = (r\cos\theta,\ r\sin\theta,\ z)$$

- Distance from the origin $|\boldsymbol{x}| = \sqrt{r^2 + z^2}$





Spherical coordinates

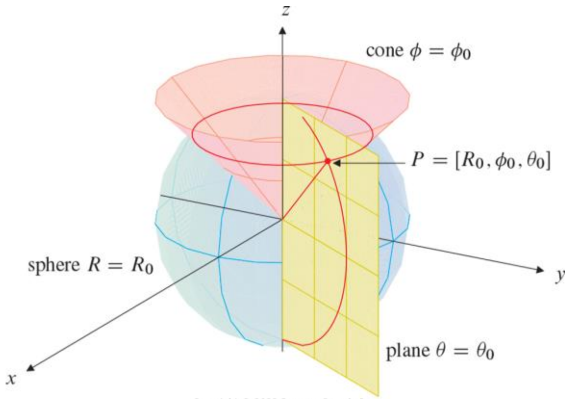
- Useful coordinate system to describe radially symmetric phenomena such Columbs law, atoms, balls, etc.

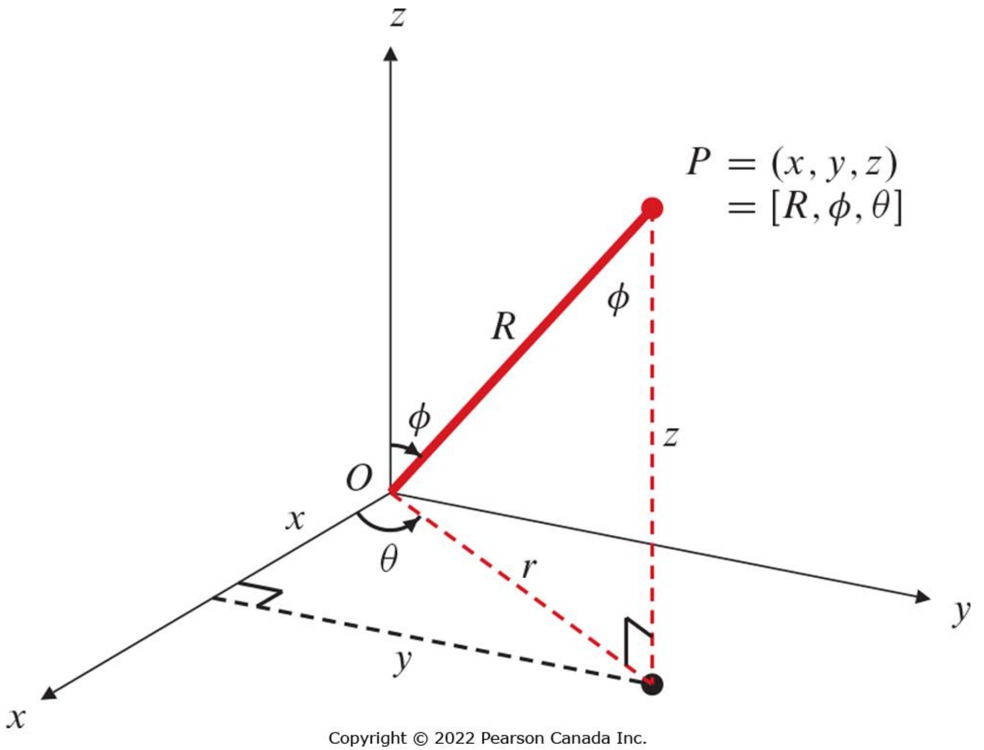
- Represented by $[R, \phi, \theta]$ where $R \geq 0, \phi \in [0, \pi], \theta \in [-\pi, \pi]$

- The coordinate in in terms of (x, y, z)

$$\boldsymbol{x} = (R\ \sin\phi\cos\theta,\ R\sin\phi\sin\theta,\ R\cos\phi)$$

- Observe that $\tan\phi = \frac{r}{z} = \frac{\sqrt{x^2+y^2}}{z},\ \tan\theta = \frac{y}{x}$





Different coordinate systems

Questions

1. What are the surfaces that correspond to

$$\theta = \frac{\pi}{4}, \phi = \frac{\pi}{3}, R = 2, r = 4?$$

2. What is the equation of an ellipsoid in cylindrical coordinate centered at (0,0,0) and having major axis lengths 6,6,4 along x, y, z axes?

Surface match